Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/16/2015
PI Name:	Lawler, John Ph.D.		
Project Title:	Redox Regulation of nNOS Translocation and Muscle Atrophy During Mechanical Unloading		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	08/24/2012	**End Date:**	08/22/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	5	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA ARC
Contact Monitor:	Taylor, Elizabeth	**Contact Phone:**	650.604.1783
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Flight Program:			
Flight Assignment:	NOTE: Extended to 8/22/2018 (previously extended in 2016 to 8/22/2017) perNSSC information (Ed., 10/6/17) NOTE: Extended to 8/22/2017 per F. Hernandez/ARC and NSSC information (Ed., 8/30/16) NOTE: Extended to 8/23/2016 per A. Chu/ARC and NSSC information (Ed., 6/16/15) NOTE: Grant number changed to NNX13AE45G, per A. Chu via NSSC (Ed., 1/29/13) NOTE: Grant number changed to NNX12AR62G, per A. Chu via NSSC (Ed., 9/28/12) NOTE: Period of performance (previously 6/27/12-6/26/15) and grant number (previously NNX12AL23G) changed per A. Chu/ARC and NSSC information (Ed., 9/12/2012)		
Key Personnel Changes/Previous PI:	No key personnel changes		
COI Name (Institution):	Fluckey, James Ph.D. (Texas A&M University)		
Grant/Contract No.:	NNX13AE45G ; NNX12AR62G (previous grant #)		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A critical limitation to astronaut performance during spaceflight missions including those to the International Space Station (ISS) and Mars is the impaired contractile ability of skeletal muscles. Skeletal muscle is a highly specialized tissue that rapidly adapts to dynamic changes in mechanical loading by modifying its ultrastructure and mass. Detection and transmittance of loading are vital to muscle integrity and protein turnover, with a nexus associated with the subsarcolemmal cytoskeleton that includes the dystrophin-glycoprotein complex (DGC) and caveolae. This interface between myocytes and the extracellular matrix confers mechanical integrity and activates cell signaling pathways that regulate protein turnover, mitochondrial, muscle fiber cross-sectional area, and fiber phenotype. Mechanical unloading experienced in the microgravity of spaceflight elicits large decrements in force generating capacity via reduction in muscle fiber cross sectional area or atrophy. While unloading-induced atrophy is a function of reduced protein synthesis coupled with increased protein degradation, activation of the transcription factors FoxO3a and nuclear factor-kappaB have drawn significant attention as mediators. Although the upstream triggers are not fully understood, translocation of neuronal nitric oxide synthase (nNOS) from the DGC and sarcolemma to the sarcoplasm is a novel signaling event that stimulates proteolysis, and thus atrophy. New clues can be gleaned from Duchenne and many of the limb-girdle muscular dystrophies, pathologies involving mutations to DGC, and other sarcolemmal proteins. Because MDs and microgravity both increase susceptibility of muscles to damage, oxidative stress, and pro-inflammatory signaling, a shared root cause is suspected: disruption of the subsarcolemmal cytoskeleton. New studies and Preliminary Data indicate the importance of NAD(P)H oxidase (NOX), a sarcolemmal source of oxidative stress, and caveolin-3 in stimulating myopathy in muscular dystrophy models. NOX may be activated upstream by angiotensin II and c-abl, a tyrosine kinase. Further, our pilot data show that unloading-induced alterations in caveolin-3 and nNOS are redox dependent and linked to atrophy and shift in muscle fiber-type. Our central hypothesis is that oxidative stress produced by NOX pathways and mitochondria directly contribute to a sarcolemmal loss of nNOS during mechanical unloading, enhancing proteolysis, and suppressing protein synthesis. The rodent hindlimb unloading model and novel fractional synthesis and degradation will be used to test our hypotheses. Specific Aim 1 will Identify NOX-dependent signaling on nNOS translocation and muscle fiber atrophy with hindlimb unloading. Specific Aim 2 will determine directly the role of mitochondrial and cytosolic oxidative stress in movement of nNOS away from the muscle cell membrane and disruption of DGC with mechanical unloading. In short, our primary objective is to identify the trigger by which skeletal muscles switch from being an anabolic to a catabolic machine during spaceflight, causing muscle atrophy and weakness. We propose that oxidative stress produced by a NOX pathway and mitochondria move the master switch protein (nNOS) away from the cell membrane, which slows protein synthesis and accelerates breakdown of protein. Our study would directly address high-priority gaps in "Recapturing a Future for Space Exploration" including the role of reactive oxygen species in protein balance and maintaining the slow contractile phenotype. Our grant proposal would also forward our understanding of (1) alterations in the subsarcolemmal cytoskeleton and membrane domains, (2) sensing and transduction of loading and gravity, (3) protein turnover, and (4) regulatory mechanisms that govern alterations in skeletal muscle, all programmatic emphases and goals of the current Space Biology NRA (NASA Research Announcement).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Bedrest, casting, and spaceflight cause significant skeletal muscle atrophy and weakness as a result of disuse. Prolonged spaceflight presents great challenges to the physical performance and health of astronauts as a result of microgravity, or reduced gravitational forces (i.e., loads). Mechanical unloading and extreme disuse during spaceflight produce rapid and dramatic skeletal muscle and weakness. Prolonged disuse also predisposes skeletal muscle to damage when landing on Mars, in an EVA (extra-vehicular activity), or returning to Earth. Furthermore, chronic disuse also contributes to and exacerbates underlying chronic pathology, including insulin resistance and cardiovascular disease. As a result crew health, safety, and completion of mission objectives are placed in jeopardy. While the pathways that cause atrophy and weakness due to disuse remain uncertain, proteins that sense changes in forces that skeletal muscles must produce may be the key. One of these proteins is nitric oxide synthase (nNOS), which is positioned near the muscle cell membrane. During spaceflight as well as aging and some types of muscular dystrophy, nNOS appears to be transported away from the cell membrane where it transforms from a muscle building to a muscle wasting agent. We will focus on oxidative stress, produced by excess free radicals produced by proteins near the cell membrane (ex. NOX) and mitochondria and how they affect nNOS transport as well as skeletal muscle atrophy. Understanding how these key membrane proteins are regulated will not only aid us in our treatment of disuse-related atrophy, but also will allow development of new therapeutics combating sedentary lifestyle and possible application to aging, Type 2 diabetes, and muscular dystrophies. Indeed, changes that are altered during spaceflight and ground models of spaceflight also appear to change with aging and diseases that cause skeletal muscle wasting and weakness. For example, loss of nNOSμ from the sarcolemma is a commonality abound spaceflight/disuse, aging, Type 2 diabetes, and Duchenne muscular dystrophy. Oxidative stress appears to be at the root of nNOSμ loss in theses muscle myopathies. Out laboratory seeks to understand the underlying redox mechanisms that leader to perturbation in nNOS function and thus myopathy.
Task Progress:	In summary, inhibition of NADPH oxidase 2 (Nox2) and upstream angiotensin II receptor 1 reduce loss of nNOS from the cell membrane, muscle atrophy, and fiber type shift. Peptide inhibition of Nox2 also protected and normalized key membrane repair proteins. The SOD/catalase mimetic EUK-134 protects anabolic signaling, suggesting that protein synthesis could be protected by targeted antioxidant approaches. Fish oil experiments are currently ongoing, but when combined with curcumin protects against unloading-induced atrophy. Meeting highlights of the year included invited symposia presentations at the NASA Human Research Program (HRP) meeting in Galveston and the American College of Sports Medicine meeting in San Diego.
Bibliography Type:	Description: (Last Updated: 11/13/2025)
Abstracts for Journals and Proceedings	Hord JM, Guzzoni V, Garcia-Villatoro EE, Ginnings Z, Lee Y, Kuczmarski JM, Lawler JM. "Loss of nNOS sarcolemmal localization and activity during hindlimb unloading is mitigated by Losartan administration." American College of Sports Medicine 62nd Annual Meeting, San Diego, CA, May 27-30, 2015. Medicine Science in Sports & Exercise. 2015 May;47(5S). , May-2015

Abstracts for Journals and Proceedings	Lawler JM, Hord JM, Guzzoni V, Lee Y, Garcia-Villatoro EE, Ginnings Z, Kuczmarski JM. "Role of Nox2 on mechanical unloading-induced skeletal muscle atrophy." American College of Sports Medicine 62nd Annual Meeting, San Diego, CA, May 27-30, 2015. Medicine Science in Sports & Exercise. 2015 May;47(5S). , May-2015
Abstracts for Journals and Proceedings	Garcia-Villatoro EE, Guzzoni V, Botchlett R, Hord JM, Lawler JM. "Effect of fish oil and curcumin on reducing unloading-induced skeletal muscle atrophy." 2015 Texas A&M Agrilife Food and Nutrition Conference, March 2015. 2015 Texas A&M Agrilife Food and Nutrition Conference, March 2015. , Mar-2015
Abstracts for Journals and Proceedings	Shimkus, KL, Lee Y, Savio EM, Wiggs, MO, Macias BR, Deaver JW, Hord JM, Lima F, Shirazi-Fard Y, Green ES, Braby LA, Hogan HA, Lawler JM, Bloomfield, SA, Fluckey JD. "Radiation Exacerbates Alterations to Skeletal Muscle Morphology Subjected to Simulated Lunar Environment." 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Abstracts for Journals and Proceedings	Lawler JM. "Spaceflight Sarcopenia: Redox Regulation of nNOSμ Translocation." 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Articles in Peer-reviewed Journals	Lawler JM, Hord JM. "CrossTalk Debate 20: The dominant mechanism causing disuse muscle atrophy is decreased protein synthesis/proteolysis. Redox Regulation of Protein Turnover During Unloading- induced Atrophy." Journal of Physiology, CrossTalk debate 20 published 15 December 2014. Comment on CrossTalk debate 20, February 2015. , Feb-2015
Articles in Peer-reviewed Journals	Kim JH, Lee Y, Kwak HB, Lawler JM. "Lifelong wheel running exercise and mild caloric restriction attenuate nuclear EndoG in the aging plantaris muscle." Experimental Gerontology. 2015 Sep;69:122-8. Epub 2015 Jun 6. http://dx.doi.org/10.1016/j.exger.2015.06.007 ; PubMed PMID: 26055450 , Sep-2015
Articles in Peer-reviewed Journals	Kwak HB, Lee Y, Kim JH, Van Remmen H, Richardson AG, Lawler JM. "MnSOD overexpression reduces fibrosis and pro-apoptotic signaling in the aging mouse heart." J Gerontol A Biol Sci Med Sci. 2015 May;70(5):533-44. http://dx.doi.org/10.1093/gerona/glu090 ; PubMed PMID: 25016531; PubMed Central PMCID: PMC4462657 , May-2015
Awards	Garcia-Villatoro E. (Erika Garcia-Villatoro) "Graduate Research Fellowship. Department of Nutrition & Food Science. Texas A&M University. September 2014. " Sep-2014